Answer **all** the questions in this Section in the spaces provided.

1 Two spheres A and B of masses 69 g and 250 g respectively are suspended by light inextensible threads from fixed points. When the spheres are at rest, they are in contact with their centres at the same horizontal level, as shown in Fig. 1.1.

The vertical distance from the fixed point of suspension to the centre of sphere A is 1.50 m.

A force F, acting at an angle θ to the vertical, is applied to sphere A, displacing it so that the thread makes an angle of 24° with the vertical, as shown in Fig. 1.2.

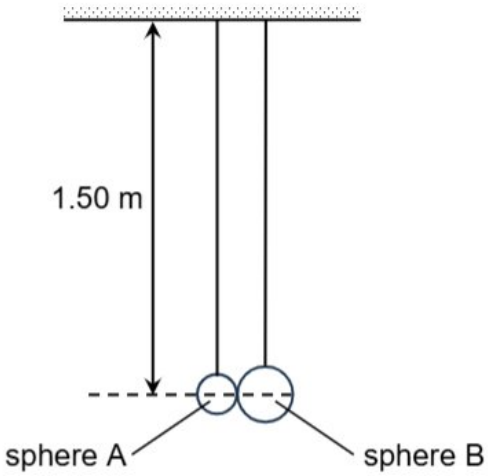


Fig. 1.1 (not to scale)

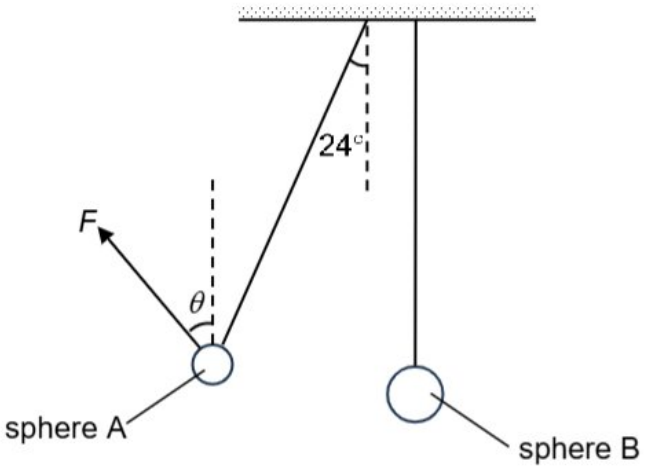


Fig. 1.2 (not to scale)

(a) (i) Calculate the magnitude of the force F required to hold sphere A in equilibrium when θ is 40°.

F = ... N [3]

(ii) By considering the vector triangle for the forces acting on sphere A, deduce the value of θ for which the force F required to maintain equilibrium is a minimum.

$\theta =$ ..... ° [1]

(b) The force F is removed and sphere A is released from rest. It subsequently collides with sphere B.

(i) Determine the work done by the gravitational force on sphere A, from the point of release to the instant just before collision.

work done by the gravitational force = ... J [1]

(ii) Hence, show that the momentum of sphere A just before collision is 0.11 kg m s^{-1}.

[1]

(c) After the collision, sphere A rebounds in the opposite direction while retaining 52% of its kinetic energy just before impact. The remainder of its kinetic energy is either transferred to sphere B or transformed into other forms of energy.

(i) Calculate the momentum of sphere B after the collision.

momentum = ... kg m s^{-1} [3]

(ii) A student states that if the collision were such that sphere A retains none of its initial kinetic energy, more kinetic energy would be transferred to sphere B.

Comment on the validity of this statement.

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[Total: 11]